Reference angles and trigonometric f[redacted]ions

1 Consider the trigonometric ratio $\sin 135\degree$.

a State which quadrant $135\degree$ is located.

b Find the measure of the positive acute angle that $135\degree$ is related to.

c Is $\sin 135\degree$ positive or negative?

d Rewrite $\sin 135\degree$ in terms of its reference angle.

2 Consider the trigonometric ratio $\cos 240\degree$.

a State which quadrant $240\degree$ is located.

b Find the measure of the positive acute angle that $240\degree$ is related to.

c Is $\cos 240\degree$ positive or negative?

d Rewrite $\cos 240\degree$ in terms of its reference angle.

3 Consider the trigonometric ratio $\tan 210\degree$.

a State which quadrant $210\degree$ is located.

b Find the measure of the positive acute angle that $210\degree$ is related to.

c Is $\tan 210\degree$ positive or negative?

d Rewrite $\tan 210\degree$ in terms of its reference angle.

4 Consider the trigonometric ratio $\sin \dfrac{7\pi}{6}$.

a State which quadrant $\dfrac{7\pi}{6}$ is located.

b Find the measure of the positive acute angle that $\dfrac{7\pi}{6}$ is related to.

c Is $\sin \dfrac{7\pi}{6}$ positive or negative?

d Rewrite $\sin \dfrac{7\pi}{6}$ in terms of its reference angle.

5 Consider the trigonometric ratio $\cos \frac{11\pi}{6}$.

a State which quadrant $\frac{11\pi}{6}$ is located.

b Find the measure of the positive acute angle that $\frac{11\pi}{6}$ is related to.

c Is $\cos \frac{11\pi}{6}$ positive or negative?

d Rewrite $\cos \frac{11\pi}{6}$ in terms of its reference angle.

6 Consider the trigonometric ratio $\tan \frac{5\pi}{4}$.

a State which quadrant $\frac{5\pi}{4}$ is located.

b Find the measure of the positive acute angle that $\frac{5\pi}{4}$ is related to.

c Is $\tan \frac{5\pi}{4}$ positive or negative?

d Rewrite $\tan \frac{5\pi}{4}$ in terms of its reference angle.

7 Consider the trigonometric ratio $\tan \frac{11\pi}{6}$.

a State which quadrant $\frac{11\pi}{6}$ is located.

b Find the measure of the positive acute angle that $\frac{11\pi}{6}$ is related to.

c Is $\tan \frac{11\pi}{6}$ positive or negative?

d Rewrite $\tan \frac{11\pi}{6}$ in terms of its reference angle.

8 Consider the trigonometric ratio $\cos \frac{7\pi}{4}$.

a State which quadrant $\frac{7\pi}{4}$ is located.

b Find the measure of the positive acute angle that $\frac{7\pi}{4}$ is related to.

c Is $\cos \frac{7\pi}{4}$ positive or negative?

d Rewrite $\cos \frac{7\pi}{4}$ in terms of its reference angle.

9 Consider the trigonometric ratio $\sin \dfrac{4\pi}{3}$.

a State which quadrant $\dfrac{4\pi}{3}$ is located.

b Find the measure of the positive acute angle that $\dfrac{4\pi}{3}$ is related to.

c Is $\sin \dfrac{4\pi}{3}$ positive or negative?

d Which of the following is $\sin \dfrac{4\pi}{3}$ equivalent to: $-\sin \dfrac{\pi}{3}$ or $\sin \dfrac{\pi}{3}$?

10 Consider the trigonometric ratio $\cos \dfrac{4\pi}{3}$.

a State which quadrant $\dfrac{4\pi}{3}$ is located.

b Find the measure of the positive acute angle that $\dfrac{4\pi}{3}$ is related to.

c Is $\cos \dfrac{4\pi}{3}$ positive or negative?

d Which of the following is $\cos \dfrac{4\pi}{3}$ equivalent to: $-\cos \dfrac{\pi}{3}$ or $\cos \dfrac{\pi}{3}$?

11 Consider the trigonometric ratio $\tan \dfrac{7\pi}{4}$.

a State which quadrant $\dfrac{7\pi}{4}$ is located.

b Find the measure of the positive acute angle that $\dfrac{7\pi}{4}$ is related to.

c Is $\tan \dfrac{7\pi}{4}$ positive or negative?

d Which of the following is $\tan \dfrac{7\pi}{4}$ equivalent to: $\tan \dfrac{\pi}{4}$ or $-\tan \dfrac{\pi}{4}$?

Exact values of trigonometric functions

12 Consider the angle $810\degree$.

a State the coordinates of the point on the unit circle that corresponds to $810\degree$.

b Find the value of $\cos 810\degree$.

c Find the value of $\sin 810\degree$.

d Determine whether the following are undefined:

i $\cot 810\degree$ ii $\csc 810\degree$ iii $\sec 810\degree$ iv $\tan 810\degree$

13 Consider the angle $(-360\degree)$.

a State the coordinates of the point on the unit circle that corresponds to $(-360\degree)$.

b Find the value of $\cos(-360\degree)$.

c Find the value of $\sin(-360\degree)$.

d Determine whether the following are undefined:

i $\cot(-360\degree)$ ii $\csc(-360\degree)$ iii $\sec(-360\degree)$ iv $\tan(-360\degree)$

14 Consider the angle $270\degree$.

a State the coordinates of the point on the unit circle that corresponds to $270\degree$.

b Find the value of $\cos 270\degree$.

c Find the value of $\sin 270\degree$.

d Determine whether the following are undefined:

i $\tan 270\degree$ ii $\cot 270\degree$ iii $\csc 270\degree$ iv $\sec 270\degree$

15 Consider the angle $(-180\degree)$.

a State the coordinates of the point on the unit circle that corresponds to $(-180\degree)$.

b Find the value of $\cos(-180\degree)$.

c Find the value of $\sin(-180\degree)$.

d Determine whether the following are undefined:

i $\sec(-180\degree)$ ii $\cot(-180\degree)$ iii $\csc(-180\degree)$ iv $\tan(-180\degree)$

16 Consider the angle 720°.

a State the coordinates of the point on the unit circle that corresponds to 720°.

b Find the value of $\cos 720^\circ$.

c Find the value of $\sin 720^\circ$.

d Determine whether the following are undefined:

i $\sec 720^\circ$ ii $\cot 720^\circ$ iii $\tan 720^\circ$ iv $\csc 720^\circ$

17 Consider the angle 1260°.

a State the coordinates of the point on the unit circle that corresponds to 1260°.

b Find the value of $\cos 1260^\circ$.

c Find the value of $\sin 1260^\circ$.

d Determine whether the following are undefined:

i $\csc 1260^\circ$ ii $\cot 1260^\circ$ iii $\tan 1260^\circ$ iv $\sec 1260^\circ$

18 Find the exact value of the following trigonometric ratios:

a $\sin 330^\circ$ b $\cos 330^\circ$ c $\tan 330^\circ$ d $\csc 330^\circ$

e $\sec 330^\circ$ f $\cot 330^\circ$ g $\sin 510^\circ$ h $\cos 510^\circ$

i $\tan 510^\circ$ j $\csc 510^\circ$ k $\sec 510^\circ$ l $\cot 510^\circ$

19 Find the exact value of the following trigonometric ratios:

a $\sin \dfrac{2\pi}{3}$ b $\cos \dfrac{2\pi}{3}$ c $\tan \dfrac{2\pi}{3}$ d $\csc \dfrac{2\pi}{3}$

e $\sec \dfrac{2\pi}{3}$ f $\cot \dfrac{2\pi}{3}$

20 For each of the following trigonometric ratios:

i Find the value of the reference angle.

ii Find the exact value of the trigonometric ratio.

a $\cos(-240^\circ)$ b $\sin 5400^\circ$ c $\cos\left(-\dfrac{4\pi}{3}\right)$

21 Find the exact values of the following:

a $\cos 0^\circ$
b $\sin 90^\circ$
c $\cos(-180^\circ)$
d $\sin 630^\circ$
e $\tan 90^\circ$
f $\tan(-630^\circ)$
g $\cot 30^\circ$
h $\csc 60^\circ$
i $\sec 30^\circ$
j $\sec 45^\circ$
k $\csc 45^\circ$
l $\cot 45^\circ$
m $\cot 270^\circ$
n $\sec 180^\circ$
o $\csc 0^\circ$
p $\cot 450^\circ$
q $\csc 900^\circ$
r $\sec(-450)^\circ$

22 Find the exact values of the following:

a $\sin \frac{\pi}{2}$
b $\sin\left(-\frac{\pi}{6}\right)$
c $\cos\left(-\frac{5\pi}{3}\right)$
d $\cos 4\pi$
e $\sin \frac{5\pi}{6}$
f $\cos \frac{3\pi}{4}$
g $\sin \frac{7\pi}{6}$
h $\cos \frac{7\pi}{6}$
i $\tan \frac{7\pi}{6}$
j $\sin \frac{5\pi}{3}$
k $\cos \frac{5\pi}{3}$
l $\tan \frac{3\pi}{4}$
m $\tan \frac{11\pi}{6}$
n $\tan \frac{3\pi}{2}$
o $\sec \frac{\pi}{2}$
p $\cot 3\pi$

23 By rewriting each ratio in terms of the related acute angle, determine the exact value of the following:

a $$\frac{\sin 120^\circ \cos 240^\circ \tan 330^\circ}{\tan(-45)^\circ}$$

b $$\frac{\left(\sin\frac{2\pi}{3}\right)\left(\cos\frac{2\pi}{3}\right)\left(\tan\frac{3\pi}{4}\right)}{\tan\left(-\frac{\pi}{4}\right)}$$

c $$\frac{\sin\left(\frac{2\pi}{3}\right)+\cos\left(\frac{5\pi}{6}\right)-\tan\left(\frac{7\pi}{4}\right)}{\cos\left(\frac{4\pi}{3}\right)}$$

d $$\frac{-\sin\left(\frac{5\pi}{3}\right)+\cos\left(\frac{7\pi}{6}\right)+\tan\left(\frac{5\pi}{3}\right)}{-\cos\left(\frac{2\pi}{3}\right)}$$